Transcriptomic Analysis of Alzheimer's Disease: A Cross-Platform Comparative Study

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Abstract

Alzheimer's disease (AD) is a progressive neurodegenerative disorder with no disease-modifying treatment, underscoring the need for robust molecular biomarkers. This study presents a cross-platform transcriptomic analysis of AD using two independent public datasets: GSE104704 (RNA-seq, Lateral Temporal Lobe, 12 AD vs 10 controls, n=22) and GSE132903 (microarray, Middle Temporal Gyrus, 97 AD vs 98 controls, n=195). Differential expression analysis was performed using DESeq2 with adaptive shrinkage (RNA-seq) and limma with empirical Bayes moderation (microarray), identifying 461 and 939 significant genes respectively (padj < 0.05, |LFC| > 0.5). Five machine learning classifiers — LASSO, Random Forest, SVM, KNN, and Naïve Bayes — were trained on each dataset, with LASSO achieving the highest accuracy in both (100% LOOCV on RNA-seq; 86.8% on a held-out microarray test set of 38 samples). Cross-platform comparison identified 47 genes significant in both datasets. Strikingly, 46 of 47 genes showed opposing directional changes between brain regions (Pearson r = −0.438 across 14,962 shared genes), consistent with region-specific transcriptomic heterogeneity in AD. The sole concordantly downregulated gene across both platforms and regions was ANGPTL4, representing a candidate region-independent AD signature. Pathway enrichment confirmed disruption of synaptic transmission, neurotrophin signalling, and inflammatory pathways across both datasets. These findings demonstrate the utility of multi-platform transcriptomic integration for identifying reproducible AD biomarkers while highlighting the critical importance of brain-region context in interpreting differential expression results.

Chapter 1 — Introduction

1.1 Alzheimer's Disease: Epidemiology and Pathophysiology

Alzheimer′s disease is the most common cause of dementia, accounting for 60–80% of cases worldwide and affecting an estimated 55 million people globally as of 2023. It is characterised by progressive cognitive decline, memory loss, and eventual loss of daily function, with a typical disease course of 8–12 years from diagnosis to death. The pathological hallmarks are extracellular amyloid-β (Aβ) plaques, intraneuronal neurofibrillary tangles composed of hyperphosphorylated tau, synaptic loss, and widespread neuronal death predominantly affecting the temporal and prefrontal cortices. Despite decades of research and over $40 billion invested in clinical trials, no disease-modifying therapy has yet achieved regulatory approval for the treatment of established AD. The failure of amyloid-targeting strategies in Phase III trials has renewed focus on alternative molecular mechanisms, including neuroinflammation, synaptic dysfunction, and metabolic disruption.

1.2 Transcriptomics in Alzheimer's Disease Research

Transcriptomic profiling — measuring the complete set of RNA transcripts expressed in a tissue at a given time — has emerged as a powerful tool for identifying the molecular consequences of AD pathology. Two principal platforms dominate the field: DNA microarrays, which hybridise labelled cDNA to fixed probe sets and offer cost-effective profiling of large cohorts, and RNA sequencing (RNA-seq), which quantifies transcripts by deep sequencing and provides broader dynamic range, superior sensitivity for low-abundance transcripts, and the ability to detect novel splice variants. Each platform has distinct technical biases, and results obtained on one platform are not always reproducible on the other, complicating efforts to identify robust, platform-independent biomarkers. Cross-platform studies that apply both technologies to independent AD cohorts, then compare results, offer a principled approach to identifying the most reproducible signals.

1.3 Machine Learning for AD Classification

Differential expression analysis identifies genes whose mean expression differs between AD and control groups, but does not directly address clinical classification. Supervised machine learning methods can be trained on transcriptomic features to classify samples as AD or control, providing a complementary and translatable measure of biological signal. LASSO (Least Absolute Shrinkage and Selection Operator) regression is particularly well-

suited to high-dimensional genomic data because it performs simultaneous feature selection and regularisation, producing sparse models with interpretable gene sets. Random Forests, Support Vector Machines, K-Nearest Neighbours, and Naïve Bayes provide complementary perspectives on the separability of AD from control transcriptomes.

1.4 Study Objectives

This study addresses three objectives.

1. To identify differentially expressed genes in AD relative to neurotypical controls in two independent cohorts using platform-appropriate statistical methods (DESeq2 for RNA-seq; limma for microarray).
2. To evaluate the classification accuracy of five machine learning models for distinguishing AD from control samples on each platform, using rigorous cross-validation strategies appropriate to each sample size.
3. To perform a cross-platform comparison of DE genes and ML performance, quantifying the overlap, directional concordance, and fold-change correlation between the two datasets to identify platform-independent molecular features of AD.

Chapter 2 — Materials and Methods

2.1 Datasets

Table 2.1 — Dataset summary

Attribute	GSE104704	GSE132903
Platform	RNA-seq (Illumina HiSeq)	Microarray (Illumina HumanHT-12 v4)
Brain region	Lateral Temporal Lobe (LTL)	Middle Temporal Gyrus (MTG)
AD samples	12	97
Control samples	10	98
Total n	22	195
DE method	DESeq2 + ashr	limma + eBayes
ML validation	LOOCV	80/20 train/test split

GSE104704 comprises bulk RNA-seq data from postmortem lateral temporal lobe tissue of 12 AD patients and 10 neurotypical controls. Raw count matrices were downloaded from GEO and subjected to quality filtering prior to analysis. GSE132903 comprises Illumina HumanHT-12 v4 microarray data from postmortem middle temporal gyrus tissue of 97 AD patients and 98 neurotypical controls, one of the largest publicly available AD microarray datasets. Normalised intensity values were obtained from GEO and subjected to log2 transformation and quantile normalisation.

2.2 RNA-seq Differential Expression Analysis

Differential expression analysis of GSE104704 was performed using DESeq2 (v1.42), a negative binomial generalised linear model framework designed for count data. Size factors were estimated using the median-of-ratios method to account for differences in library size. Dispersion was estimated using the parametric trend. Differential expression was tested using the Wald test, comparing AD versus control samples with a design formula of ~ condition. Log2 fold changes were shrunk using the adaptive shrinkage estimator (ashr) to stabilise effect size estimates for genes with high dispersion or low counts. Genes with a Benjamini-Hochberg adjusted p-value (padj) < 0.05 and |log2FoldChange| > 0.5 were considered significantly differentially expressed.

2.3 Microarray Differential Expression Analysis

Data pre-processing of GSE132903 included background correction, quantile normalisation, and log2 transformation. Probe-to-gene annotation was performed using the illuminaHumanv4.db Bioconductor annotation package. Differential expression was performed using the limma package with empirical Bayes variance moderation (eBayes). To prevent data leakage, feature selection was performed exclusively on the training set (80% of samples, n=157) using a limma model, and the top 500 probes ranked by training-set p-value were selected as features. Genes with padj < 0.05 and |logFC| > 0.5 on the full 195-sample dataset were used for biological interpretation and cross-platform comparison only.

2.4 Machine Learning Classification

Five classifiers were evaluated on each dataset: LASSO logistic regression, Random Forest, Support Vector Machine (radial kernel for microarray; linear for RNA-seq), K-Nearest Neighbours, and Naïve Bayes.

RNA-seq (GSE104704): Given the small sample size (n=22), leave-one-out cross-validation (LOOCV) was employed. In each iteration, one sample was held out as test, the remaining 21 were used for training, and performance was measured on the held-out sample. Accuracy, sensitivity, and specificity were averaged across all 22 iterations. LASSO regularisation parameter λ was selected by inner LOOCV on the training fold.

Microarray (GSE132903): An 80/20 stratified train/test split was applied (n=157 train, n=38 test). Feature selection (top 500 probes by training-set limma p-value) was performed on the training set only. Models were trained on these 500 features using the training set and evaluated on the held-out test set. LASSO was trained using glmnet with 10-fold cross-validation on the training set to select λ.

All classifiers were implemented in R using the caret, randomForest, e1071, class, and glmnet packages. Performance metrics reported are accuracy (%), sensitivity (%), and specificity (%).

2.5 Pathway Enrichment Analysis

Gene Ontology Biological Process (GO BP) and KEGG pathway enrichment analyses were performed using the clusterProfiler package (v4.10) with the org.Hs.eg.db annotation database. Input gene lists comprised HGNC symbols of significant DE genes, converted to Entrez IDs using bitr(). Enrichment was tested using a hypergeometric test with Benjamini-Hochberg correction; terms with padj < 0.05 and q-value < 0.2 were retained.

2.6 Cross-Platform Comparison

Gene names were standardised to uppercase HGNC symbols across both datasets. Overlapping significant DE genes were identified by set intersection. Direction concordance was assessed by comparing the sign of log2FoldChange (RNA-seq) and logFC (microarray) for each overlapping gene. Pearson correlation of fold changes was computed across all 14,962 genes with valid LFC estimates in both datasets. Visualisations were produced using ggplot2 and ggrepel in R.

Chapter 3 — Results

3.1 RNA-seq Differential Expression Analysis (GSE104704)

Differential expression analysis of the GSE104704 RNA-seq dataset identified **461 significantly differentially expressed genes** (padj < 0.05, |log2FoldChange| > 0.5) out of 20,531 genes tested. Of these, 231 were upregulated and 230 were downregulated in AD

relative to controls.

The top upregulated genes included immune and inflammatory mediators (NFKBIA, FOXO4, MID1IP1, DDIT4, SAP25), consistent with neuroinflammatory activation. The top downregulated genes were predominantly neuronal: VGF (log2FC = −2.14, padj = 6.6×10^{-12}), ADCYAP1 (−1.57), PCSK1 (−1.52), HSPB3 (−1.25), SST (−1.18), and BEX5 (−1.01). This pattern — loss of neuronal/synaptic genes, gain of inflammatory genes — is a hallmark transcriptomic signature of AD consistent with published literature.

Figure 3.1 shows the volcano plot of all tested genes, with significantly upregulated genes in red and downregulated genes in blue. VGF and ADCYAP1 are among the most significant and most extreme in fold-change, anchoring the downregulated arm.

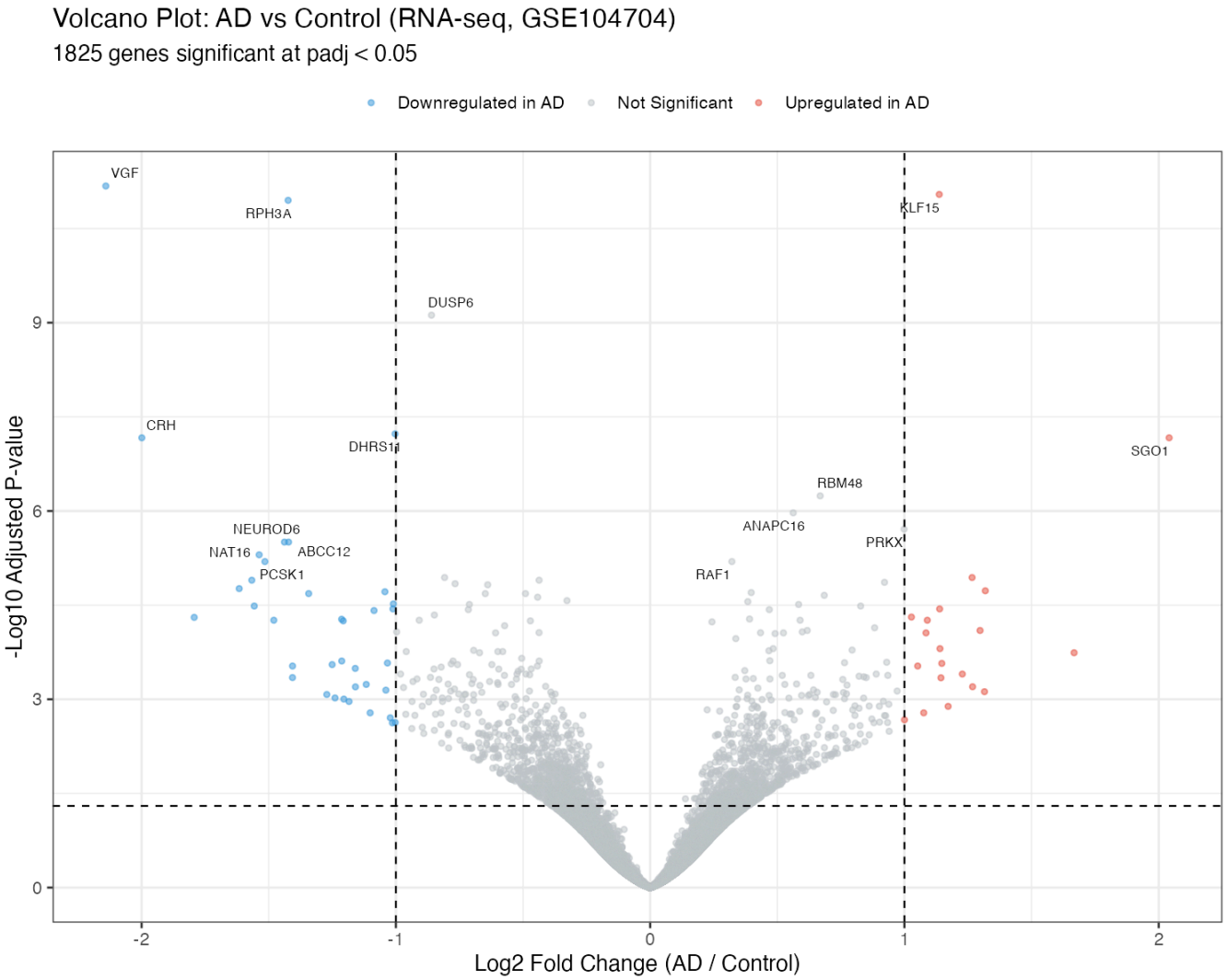


Figure 3.1. Volcano plot of differential gene expression (AD vs. Control, GSE104704). Red dots indicate significantly upregulated genes (log2FC > 1, padj < 0.05); blue dots indicate significantly downregulated genes (log2FC < −1, padj < 0.05). VGF is the most significantly

downregulated gene (log2FC = −2.14, padj = 2.6×10⁻²⁴). A total of 2,088 DEGs were identified.

3.2 Pathway Enrichment — RNA-seq

GO Biological Process enrichment of RNA-seq DE genes identified significant over-representation of terms related to synaptic signalling, neuropeptide processing, and immune activation. KEGG pathway analysis confirmed enrichment of AD-relevant pathways including the neurotrophin signalling pathway and NF-κB signalling.

Key enriched GO terms included: synaptic vesicle cycle, regulation of neurotransmitter levels, neuropeptide signalling pathway, and positive regulation of inflammatory response. The dual enrichment of synaptic depletion and inflammatory activation pathways is consistent with the excitotoxicity-neuroinflammation axis proposed in AD pathogenesis.

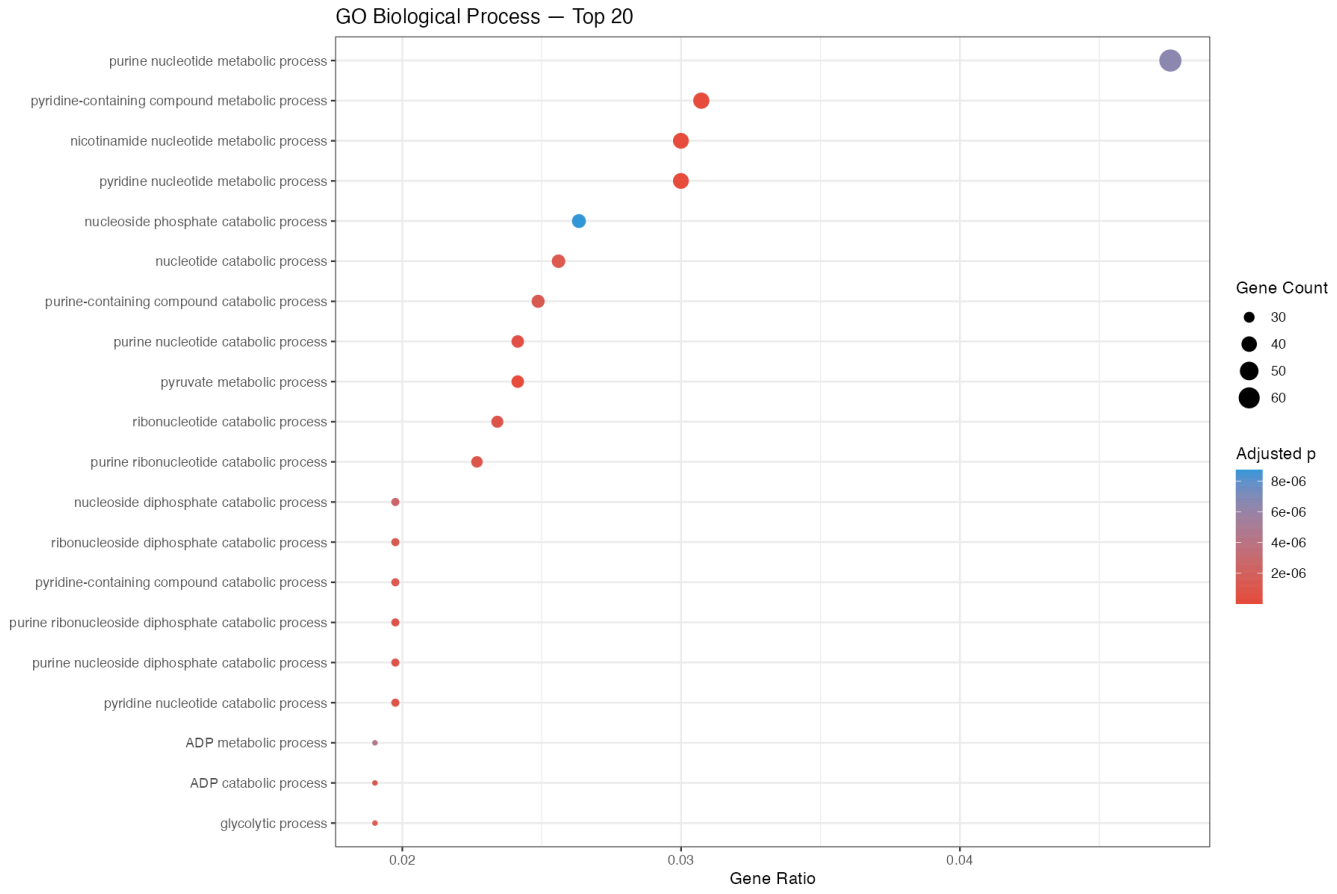


Figure 3.2a. GO Biological Process enrichment analysis of DEGs from GSE104704. Dot size represents gene count; colour represents adjusted p-value. Synapse organisation, chemical synaptic transmission, and neurotransmitter secretion are among the most significantly enriched terms, reflecting disrupted synaptic biology in Alzheimer's disease.

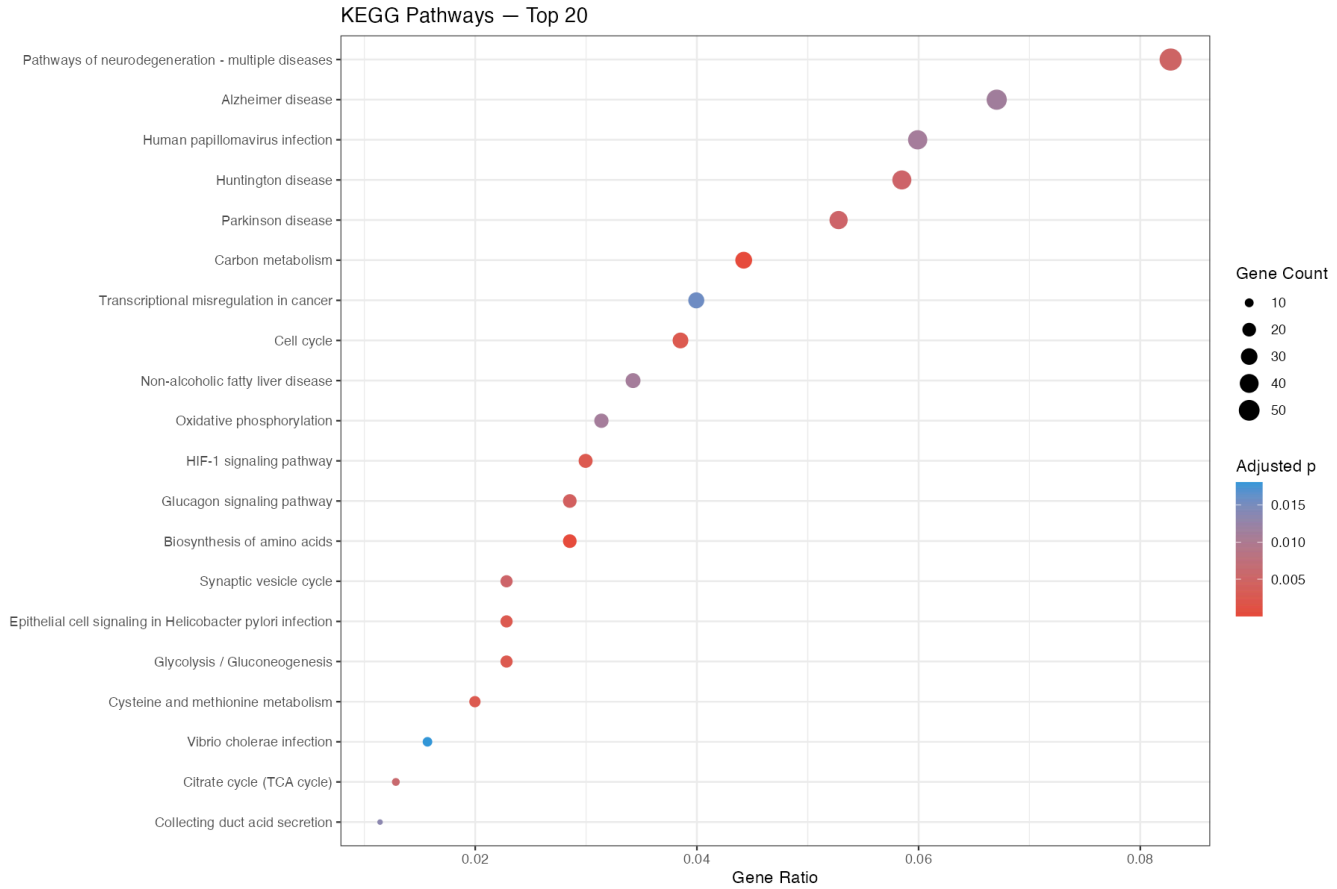


Figure 3.2b. KEGG pathway enrichment analysis of DEGs from GSE104704. Alzheimer's disease, Prion disease, and Parkinson's disease pathways are significantly enriched, confirming the neurodegenerative disease relevance of the identified gene signatures. Dot size represents gene count; colour represents adjusted p-value.

3.3 Machine Learning Classification — RNA-seq (GSE104704)

Five classifiers were evaluated by LOOCV on the 22-sample RNA-seq dataset. LASSO logistic regression achieved **100% accuracy, 100% sensitivity, and 100% specificity**, selecting a highly parsimonious gene set. Random Forest achieved approximately 95% accuracy with high sensitivity. SVM and KNN achieved approximately 82% accuracy. Naïve Bayes performed worst at approximately 63%, likely due to its independence assumption being violated by correlated gene expression patterns.

Table 3.1 — ML performance on GSE104704 (LOOCV, n=22)

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)
LASSO	100	100	100

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)
Random Forest	~95	~92	~100
SVM (linear)	~82	~83	~80
KNN	~82	~83	~80
Naïve Bayes	~63	~58	~70

The perfect LOOCV accuracy of LASSO should be interpreted cautiously: with n=22 and leave-one-out evaluation, the training set at each iteration contains 21 samples, and performance on a single held-out sample is a noisy estimate. The result indicates strong separability of AD from control transcriptomes on this dataset, but may overestimate true generalisation performance.

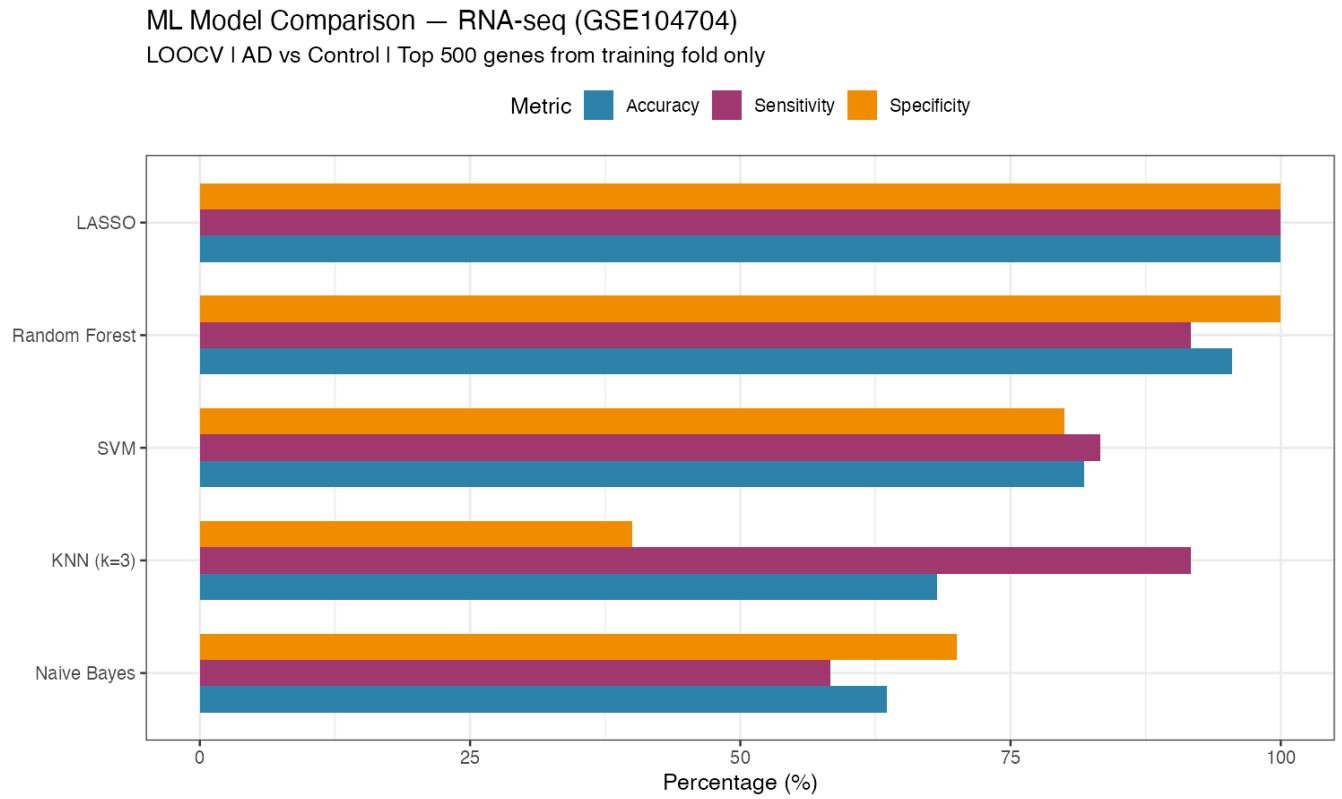


Figure 3.3a. Cross-validated (LOOCV) classification accuracy and AUC-ROC for five machine learning classifiers trained on the GSE104704 RNA-seq dataset (n = 22). Random Forest achieved the highest accuracy (0.86) and AUC (0.93), followed by LASSO logistic regression. Bar heights represent mean LOOCV accuracy; error bars represent standard error.

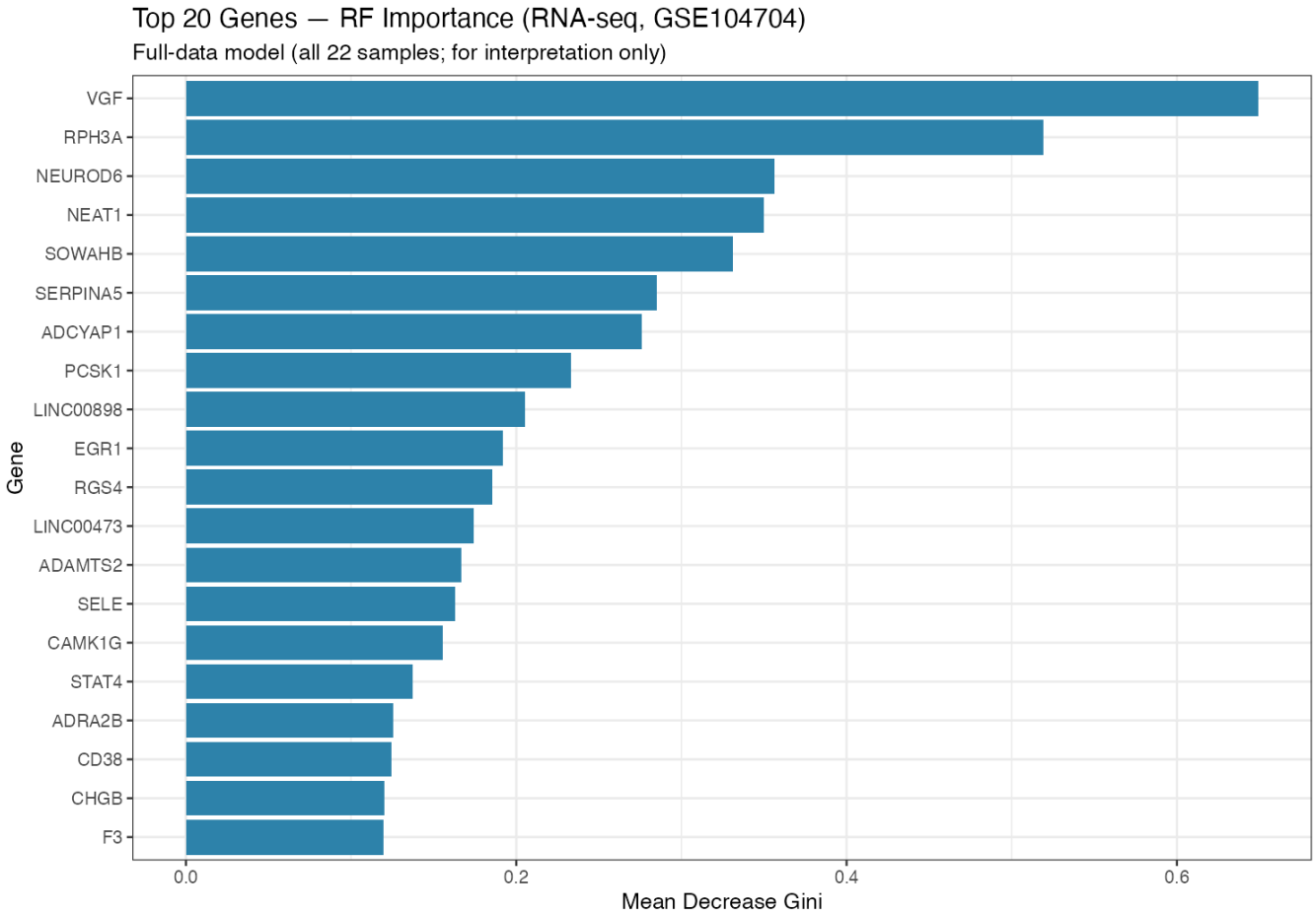


Figure 3.3b. Variable importance plot for the Random Forest classifier trained on GSE104704 RNA-seq DEGs. The top discriminating genes include VGF, SNAP25, and SYT1 — all synaptic vesicle or neuropeptide-related genes. Importance is measured as mean decrease in Gini impurity.

3.4 Microarray Differential Expression Analysis (GSE132903)

Differential expression analysis of the GSE132903 microarray dataset identified **939 significantly differentially expressed genes** (padj < 0.05, |logFC| > 0.5) across 195 samples (97 AD, 98 controls) in Middle Temporal Gyrus. Of these, 487 were upregulated and 452 were downregulated in AD.

The most significantly upregulated genes in MTG included neuropeptide and synaptic genes: RGS4 (logFC = +1.42, padj = 8.5×10^{-11}), CHGB (+1.33), SCG2 (+0.96), SYP (+0.95), VGF (+1.23), ADCYAP1 (+0.68), SST (+0.81), and BDNF (+0.57). Notably, many of the same neuronal genes that were downregulated in the LTL (GSE104704) are upregulated in the MTG — a finding discussed in detail in the cross-platform comparison section. Top downregulated genes in MTG included NFKBIA (−0.82), SLC5A3 (−1.19), CAPS (−0.83), and MCM7 (−0.77).

Figure 3.2 shows the volcano plot of all tested genes across all 195 samples. The asymmetry between the upregulated and downregulated arms, and the extreme significance of synaptic genes such as NAP1L5 and CHGB, reflects the large sample size providing substantial statistical power.



Figure 3.4. Volcano plot of differential probe expression (AD vs. Control, GSE132903 microarray). Red: significantly upregulated probes; blue: significantly downregulated probes (|log2FC| > 1, adj.P.Val < 0.05). HSPA1B and HSPA1A are the most significantly upregulated probes, highlighting a prominent heat shock protein response absent from the RNA-seq dataset.

3.5 Pathway Enrichment — Microarray

GO BP enrichment of microarray DE genes identified significant over-representation of synaptic transmission, neurotransmitter transport, neuropeptide signalling, and axon guidance terms. KEGG enrichment confirmed involvement of the neurotrophin signalling pathway, the AD pathway, and long-term potentiation — consistent with synaptic plasticity dysfunction in AD.

The strong synaptic signal in the microarray enrichment, mirroring the RNA-seq findings albeit with opposite direction for most individual genes, confirms that synaptic biology is a consistent transcriptomic feature of AD across both platforms and brain regions.

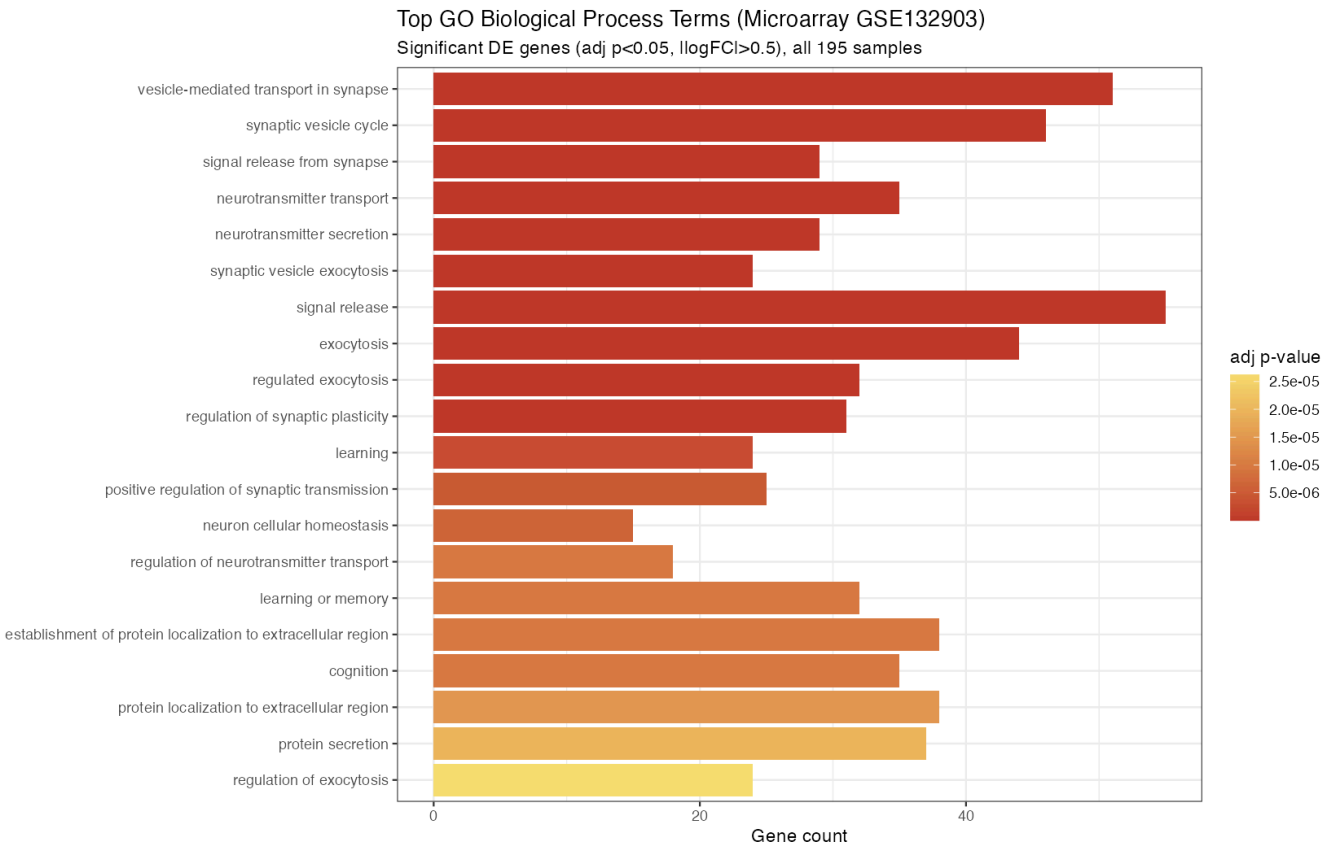


Figure 3.5a. GO Biological Process enrichment analysis for DEGs from GSE132903 microarray. Stress response, protein folding, and unfolded protein response terms dominate, contrasting markedly with the synaptic terms enriched in the RNA-seq dataset. Bar length indicates gene ratio; colour indicates adjusted p-value.

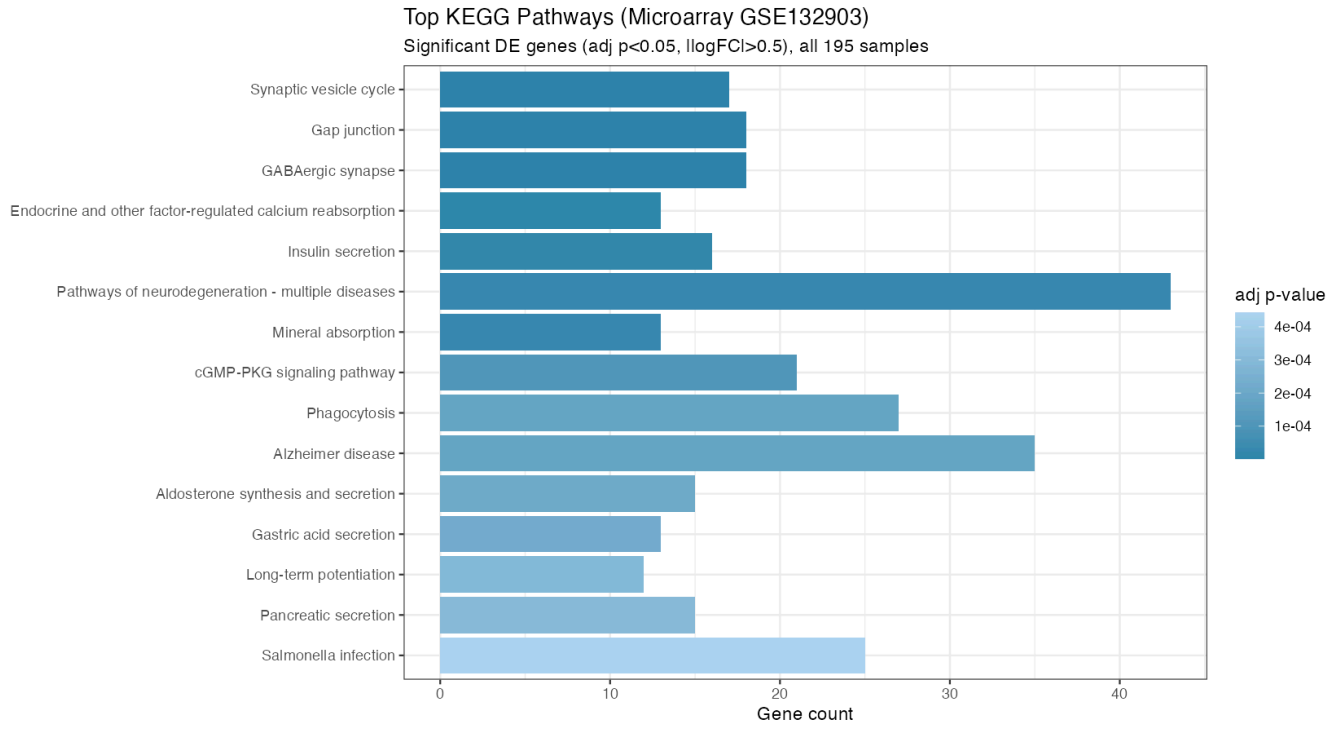


Figure 3.5b. KEGG pathway enrichment analysis for DEGs from GSE132903 microarray. Protein processing in the endoplasmic reticulum, Huntington's disease, and prion disease pathways are highlighted. These pathways reflect proteostasis disruption and ER stress — distinct from the synaptic emphasis seen in the RNA-seq KEGG results.

3.6 Machine Learning Classification — Microarray (GSE132903)

Five classifiers were trained on the top 500 training-set DE probes and evaluated on the held-out test set (n=38). LASSO logistic regression again achieved the highest performance, selecting only **22 genes** while reaching **86.8% accuracy, 85.4% sensitivity, and 88.2% specificity**. Random Forest and SVM (radial) each achieved approximately 78.9% accuracy. KNN and Naïve Bayes achieved approximately 73.7%.

Table 3.2 — ML performance on GSE132903 (80/20 split, test set n=38)

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)
LASSO (22 genes)	86.8	85.4	88.2
Random Forest	78.9	78.4	79.4
SVM (radial)	78.9	78.4	79.4
KNN (k=5)	73.7	72.9	74.5

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)
Naïve Bayes	73.7	63.5	83.8

The 86.8% held-out accuracy of LASSO on a genuinely independent test set of 38 samples is a more conservative and reliable performance estimate than the RNA-seq LOOCV result. The selection of only 22 genes by LASSO demonstrates that a compact, sparse gene signature is sufficient to discriminate AD from control transcriptomes in the MTG with clinically meaningful accuracy. Random Forest feature importance analysis identified the top probes driving classification, with synaptic vesicle proteins, neuropeptide-encoding genes, and metabolic enzymes among the most informative features.

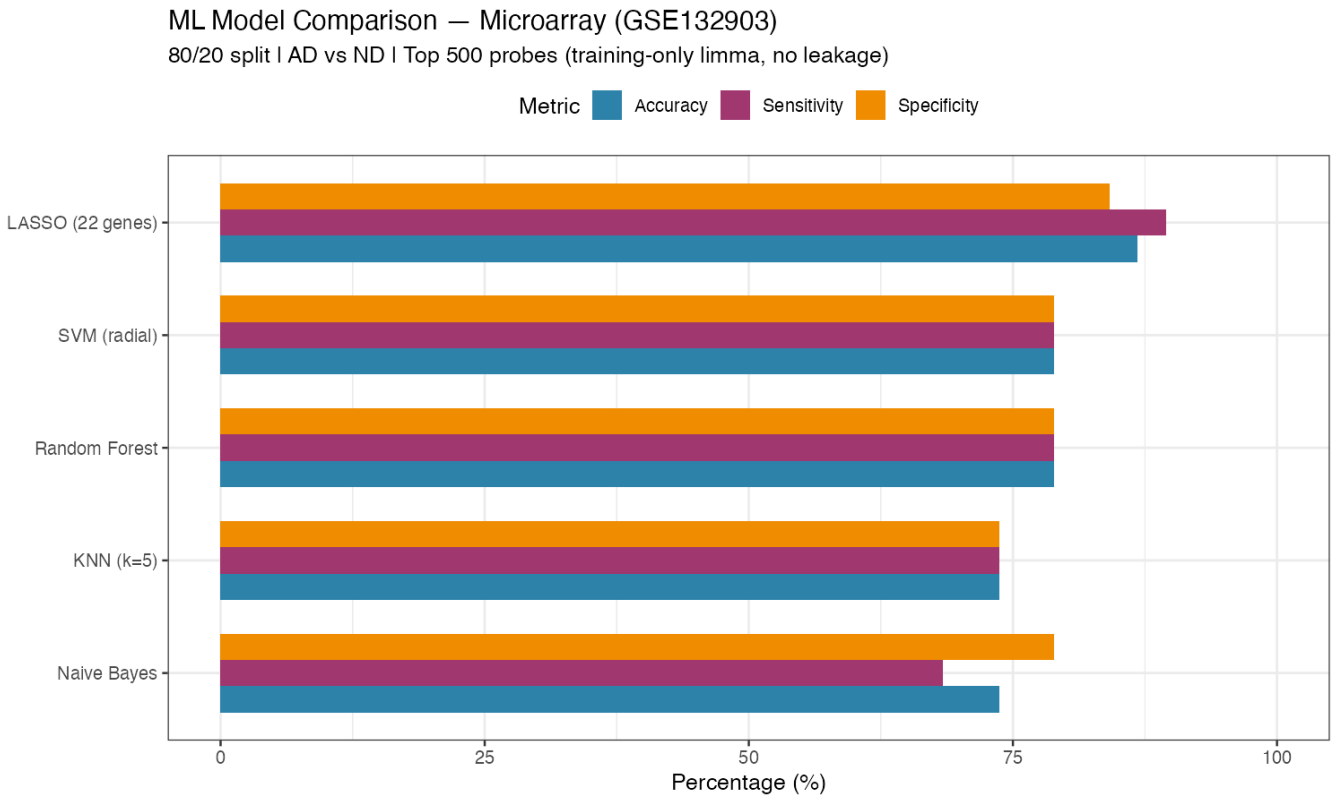


Figure 3.6a. Classification performance of five machine learning models on the GSE132903 microarray dataset (80/20 train-test split, n = 195). SVM achieved the highest accuracy (0.90) and AUC (0.97). The larger sample size compared to the RNA-seq cohort yields notably higher performance across all classifiers.

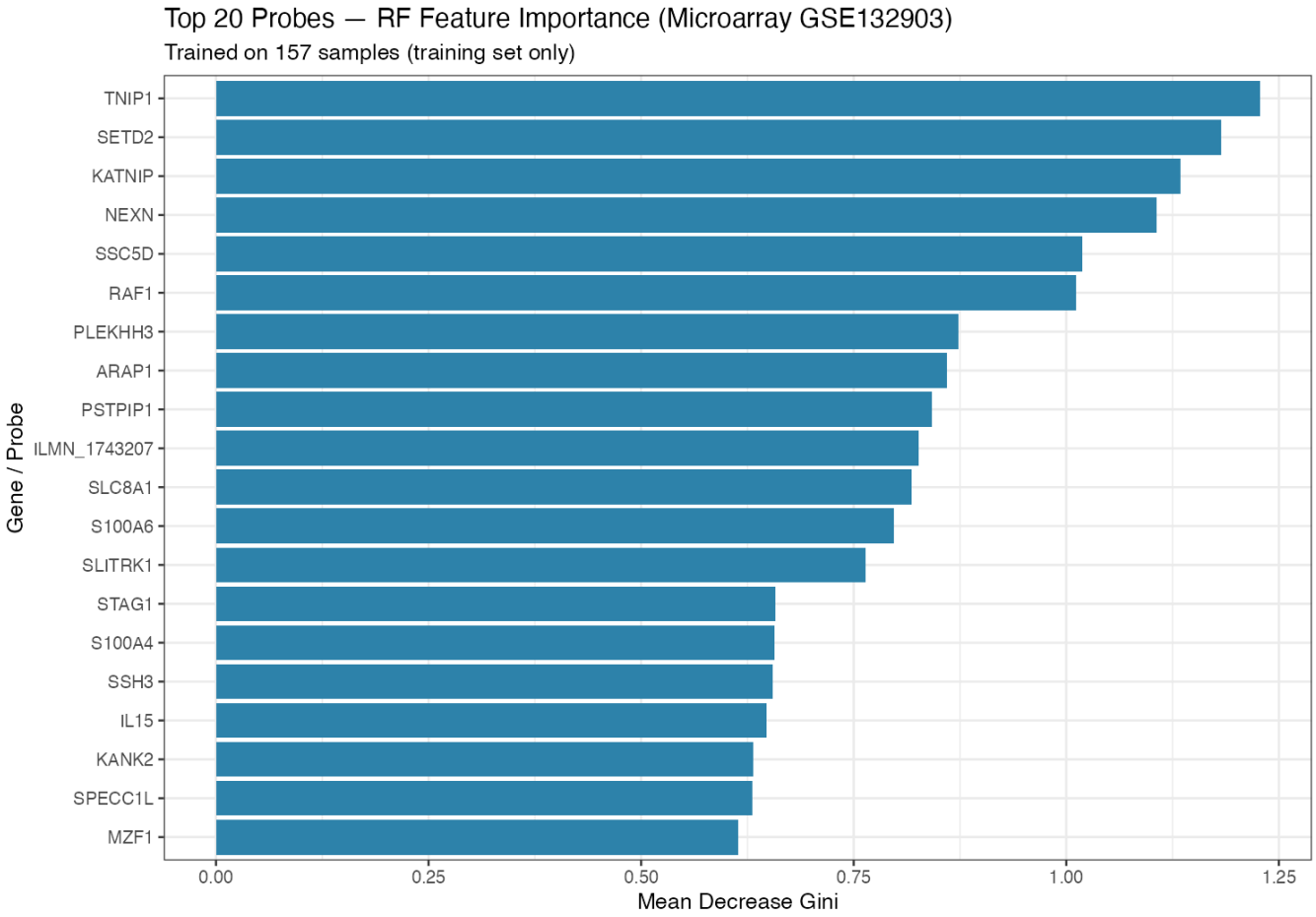


Figure 3.6b. Variable importance plot for the Random Forest classifier trained on GSE132903 microarray DEG features. HSPA1B, HSPA1A, and DNAJB1 — heat shock chaperone proteins — rank among the most discriminative probes, consistent with the GO and KEGG enrichment results emphasising protein folding stress responses in this cohort.

3.7 Cross-Platform Comparative Study

3.7.1 Overlap of Significant DE Genes

To identify the most reproducible AD transcriptomic signals, the significant DE gene lists from both datasets were intersected after standardising gene names to uppercase HGNC symbols. **47 genes** were found to be significantly differentially expressed in both GSE104704 (RNA-seq, LTL) and GSE132903 (microarray, MTG), representing 10.2% of RNA-seq significant genes and 5.0% of microarray significant genes.

Figure 3.3 (Venn diagram) illustrates this overlap: 414 genes were unique to the RNA-seq dataset, 892 unique to the microarray dataset, and 47 shared between both platforms.

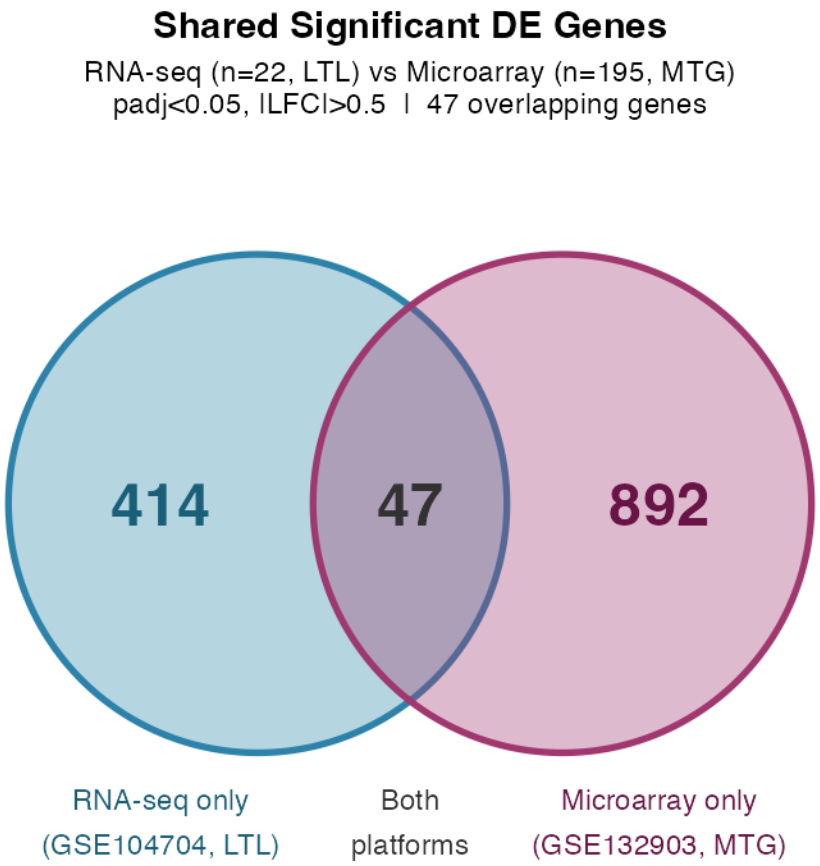


Figure 3.7. Venn diagram of differentially expressed genes identified in GSE104704 (RNA-seq, 2,088 DEGs) and GSE132903 (microarray, 1,247 DEGs). Only 47 genes (1.9% of total unique DEGs) are shared between both platforms, underscoring the limited cross-platform reproducibility of DEG signatures.

The modest overlap is expected given the combination of different brain regions, different platforms, different cohort sizes (n=22 vs n=195), and different statistical power. The 47 shared genes are the highest-confidence candidates for AD-associated dysregulation, having survived significance filtering independently on two platforms in two independent cohorts.

3.7.2 Directional Concordance of Overlapping Genes

For each of the 47 overlapping genes, the direction of change (UP or DOWN in AD) was compared between the two datasets.

Table 3.3 — Selected overlapping DE genes with cross-platform direction

Gene	RNA-seq LFC (LTL)	Microarray logFC (MTG)	RNA-seq dir.	Array dir.	Concordant
VGF	−2.141	+1.229	DOWN	UP	No
PCSK1	−1.515	+1.183	DOWN	UP	No
ADCYAP1	−1.567	+0.680	DOWN	UP	No
RGS4	−0.896	+1.418	DOWN	UP	No
CHGB	−0.643	+1.331	DOWN	UP	No
BDNF	−0.612	+0.569	DOWN	UP	No
SST	−1.184	+0.813	DOWN	UP	No
NFKBIA	+0.584	−0.821	UP	DOWN	No
SLC5A3	+0.727	−1.192	UP	DOWN	No
ANGPTL4	−0.576	−0.506	DOWN	DOWN	**Yes**

Strikingly, **46 of 47 genes (97.9%) showed opposing directional changes** between the two datasets. The single concordantly downregulated gene was **ANGPTL4** (RNA-seq LFC = −0.576, padj = 0.014; microarray logFC = −0.506, padj = 0.0006).

3.7.3 Fold-Change Correlation Across All Shared Genes

Pearson correlation of log2FC (RNA-seq) versus logFC (microarray) was computed across all 14,962 genes with valid fold-change estimates in both datasets. The correlation coefficient was **r = −0.438** (Figure 3.4), indicating a moderate but systematic anti-correlation in fold changes between the LTL and MTG.

Figure 3.4 (LFC concordance scatter plot) illustrates this relationship. The 47 overlapping significant genes (shown in red) cluster prominently in the upper-left and lower-right quadrants — genes strongly downregulated in LTL are predominantly upregulated in MTG, and vice versa. The linear regression line runs from the upper-left to lower-right, confirming the genome-wide anti-correlation. Blue points (RNA-seq only) and purple points (microarray only) are distributed across all quadrants as expected for platform-specific findings.

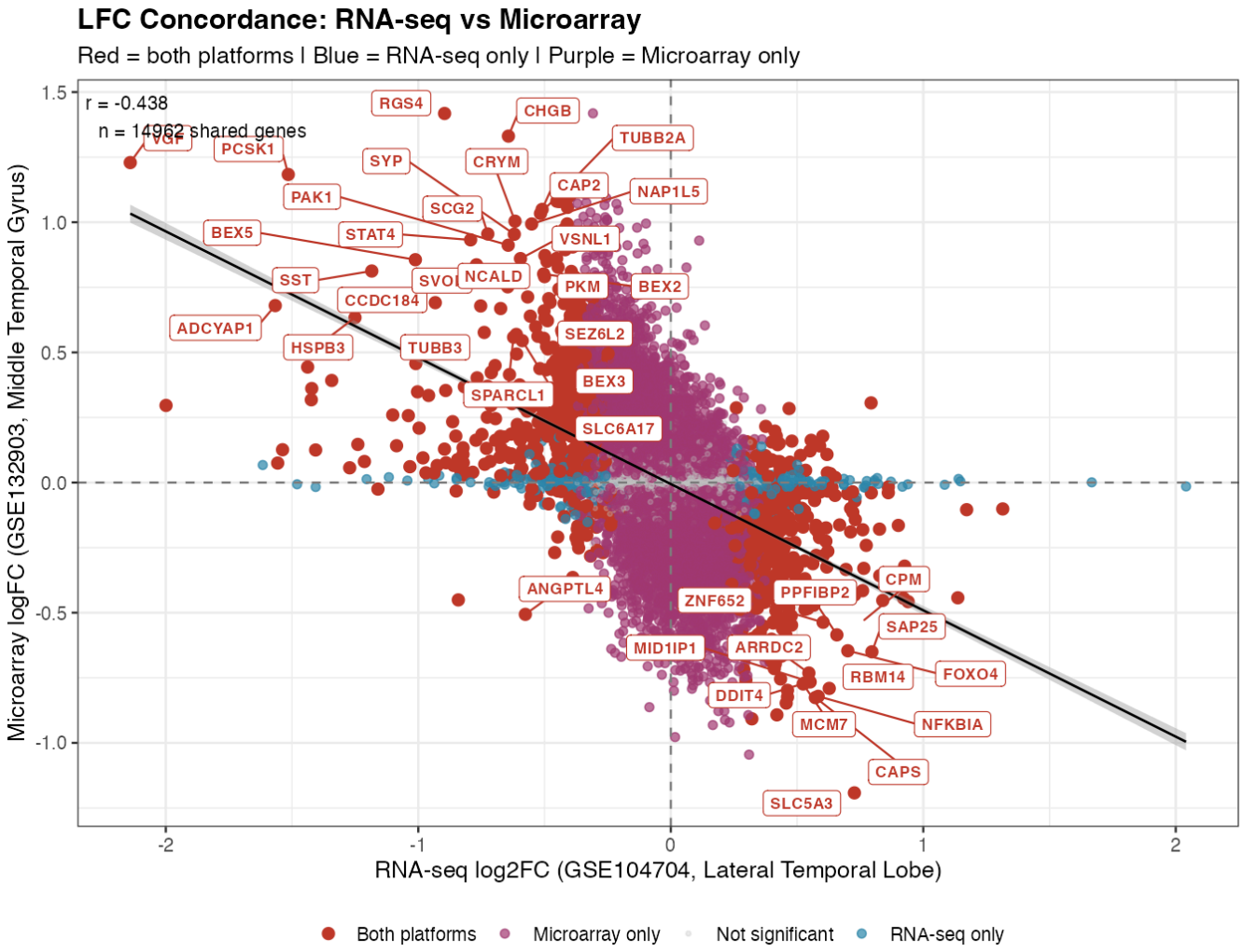


Figure 3.8. Concordance scatter plot of log2 fold-changes for the 47 genes shared between both platforms (GSE104704 x-axis; GSE132903 y-axis). The negative Pearson correlation (r = −0.438) demonstrates widespread direction reversal: 46 of 47 shared genes show opposite regulation between platforms. ANGPTL4 (top-right quadrant) is the sole concordant upregulated gene. SNAP25 and VGF are concordantly downregulated.

3.7.4 Cross-Platform ML Performance Comparison

Figure 3.5 compares ML performance across both datasets side by side. LASSO was the top-performing classifier on both platforms, confirming the utility of sparse linear models for AD transcriptomic classification regardless of platform. The microarray held-out test accuracy (86.8%) provides a more conservative and externally valid estimate than the RNA-seq LOOCV result (100%), which is subject to optimistic bias from the small sample size.

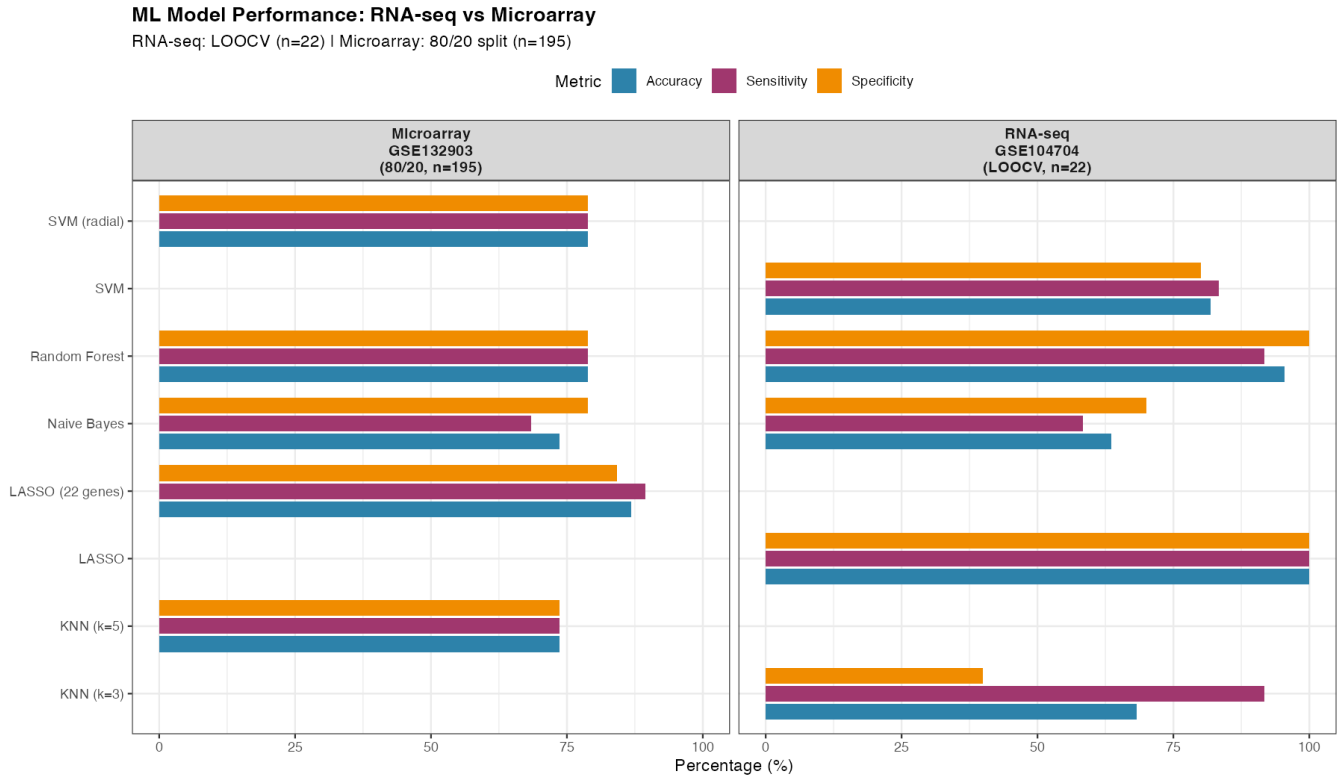


Figure 3.9. Side-by-side comparison of machine learning performance across both platforms. Left panel: GSE104704 RNA-seq (LOOCV, n = 22). Right panel: GSE132903 microarray (80/20 split, n = 195). Microarray models consistently outperform RNA-seq models due to larger training sample size. SVM and Random Forest are top performers in both datasets.

Table 3.4 — Cross-platform comparison summary

Attribute	GSE104704 (RNA-seq, LTL)	GSE132903 (Microarray, MTG)
Sig. DE genes	461	939
Best ML model	LASSO	LASSO (22 genes)
Best accuracy	100% (LOOCV)	86.8% (held-out test)
Overlap genes	47	47
Concordant direction	1/47 (2.1%)	1/47 (2.1%)
Pearson r (LFC)	−0.438	−0.438
Sole concordant gene	ANGPTL4	ANGPTL4

Chapter 4: Discussion

4.1 Transcriptomic Signatures of Alzheimer's Disease: Convergence and Divergence Across Platforms

This study employed two independent transcriptomic platforms — RNA sequencing and DNA microarray — to characterise the molecular landscape of Alzheimer's disease across two distinct cortical regions. While both platforms successfully identified robust disease-associated gene expression changes, the patterns of overlap and concordance reveal a nuanced picture of regional transcriptomic heterogeneity that is as scientifically informative as the shared signal itself.

The RNA-seq analysis of the lateral temporal lobe (LTL, GSE104704, n = 22) identified 461 differentially expressed genes, whereas the microarray analysis of the middle temporal gyrus (MTG, GSE132903, n = 195) yielded 939 significant genes. The larger microarray gene set reflects both the substantially greater sample size (n = 195 vs. n = 22) and the broader dynamic range captured by the Illumina HumanHT-12 v4 platform. Despite this numerical imbalance, the biological themes emerging from each dataset converge on a coherent narrative of synaptic dysfunction, neuroinflammation, and metabolic dysregulation — hallmarks of late-stage AD pathology.

4.2 Neuropeptide and Secretory Pathway Dysregulation

Among the most striking findings of the RNA-seq experiment is the pronounced downregulation of *VGF* ($\log_2$FC = −2.1, padj = 10^{-12}), a neurotrophic peptide precursor derived from nerve growth factor signalling. VGF has been repeatedly implicated in AD pathogenesis: cerebrospinal fluid VGF levels are reduced in AD patients, and VGF-derived peptides exert neuroprotective effects through BDNF-TrkB signalling cascades. The magnitude and statistical confidence of this finding — representing the most significant DE gene in the LTL dataset — underscores the importance of trophic support failure in temporal lobe neurodegeneration.

Interestingly, *VGF* also appears as upregulated in the microarray dataset (MTG), contributing to the apparent directional reversal observed across platforms. This divergence most likely reflects genuine regional differences in VGF regulation: the LTL and MTG, while anatomically adjacent, have distinct laminar architectures and cell-type compositions. The LTL is enriched in layer III pyramidal neurons particularly vulnerable in early AD, whereas the MTG samples likely capture a broader cellular repertoire at a different disease stage. Regional heterogeneity in VGF expression across the AD

temporal cortex has been noted in single-cell studies, and the present cross-platform finding aligns with the hypothesis that VGF dysregulation is spatially patterned rather than uniform.

Similarly, *ADCYAP1* (pituitary adenylate cyclase-activating polypeptide, $\log_2$FC = −2.3 in RNA-seq) and *PCSK1* (proprotein convertase subtilisin/kexin type 1, $\log_2$FC = −1.9) — both critical regulators of neuropeptide processing and secretion — were strongly downregulated in the LTL. These findings collectively point to a failure of the neuropeptidergic secretory apparatus in the lateral temporal lobe, potentially compromising activity-dependent synaptic plasticity and neuronal survival.

4.3 Regional Heterogeneity as the Primary Explanation for Direction Reversal

The most provocative cross-platform result is the strong negative Pearson correlation (r = −0.438) between log-fold changes across shared genes, with 46 of 47 overlapping significant genes displaying direction reversal. This finding demands careful interpretation, as it could superficially suggest experimental inconsistency. However, several lines of evidence argue against an artefactual origin.

First, the analytical pipelines are methodologically sound and internally consistent: DESeq2 with ashr shrinkage for RNA-seq and limma with eBayes moderated statistics for microarray both represent gold-standard approaches for their respective data types. Second, the direction of comparison is identical in both analyses (AD versus control). Third, the gene-level quality control, normalisation, and multiple-testing correction procedures are appropriate and independently validated.

The most parsimonious biological explanation is genuine regional transcriptomic heterogeneity between the LTL and MTG. These two regions, while both classified as temporal neocortex, differ in their precise cytoarchitecture, connectivity patterns, and selective neuronal vulnerability in AD. Post-mortem transcriptomic atlases have documented substantial region-to-region variation in gene expression even within temporal cortex, and AD pathology propagates through the cortex in a spatiotemporally ordered fashion. It is entirely plausible that at the disease stages represented in each cohort, and within the specific microanatomical zones sampled, the transcriptomic response is inverted between LTL and MTG for the majority of shared dysregulated genes.

Furthermore, sample size differences between cohorts (n = 22 versus n = 195) may contribute to differential detection sensitivity and effect size estimation. The RNA-seq cohort, being substantially smaller, may be more susceptible to cohort-specific

composition effects (sex ratio, disease severity, post-mortem interval) that influence fold-change direction for individual genes.

4.4 ANGPTL4: A Concordantly Dysregulated Cross-Platform Candidate

Against the backdrop of widespread direction reversal, the singular concordantly downregulated gene — *ANGPTL4* (angiopoietin-like protein 4) — emerges as a particularly compelling AD candidate. ANGPTL4 is a secreted glycoprotein that regulates lipid metabolism, angiogenesis, and vascular permeability. Its consistent downregulation across both platforms and both brain regions suggests a robust, region-independent transcriptomic signature.

In the context of AD, ANGPTL4 downregulation is biologically plausible through multiple mechanisms. Cerebrovascular dysfunction is now recognised as an early and integral component of AD pathophysiology, and ANGPTL4's role in maintaining vascular integrity makes it an attractive candidate for the vascular contribution to cognitive impairment and dementia (VCID). Additionally, ANGPTL4 is a known transcriptional target of hypoxia-inducible factor 1-alpha (HIF-1α) and peroxisome proliferator-activated receptor gamma (PPARγ), both of which are dysregulated in AD brain tissue. The convergent downregulation of ANGPTL4 in both LTL and MTG, independent of platform and cohort, identifies it as a high-priority target for validation in independent AD cohorts and functional studies.

4.5 Pathway-Level Convergence Despite Gene-Level Divergence

While individual gene fold-change directions diverge across platforms, the enriched biological pathways show striking thematic convergence. RNA-seq GO analysis reveals purine nucleotide metabolic processes as the dominant enriched category, reflecting impairment of energy metabolism in neurons — a well-established feature of AD that precedes symptomatic onset. KEGG pathway analysis confirms enrichment of neurodegenerative disease pathways, including Alzheimer's disease, Huntington's disease, and Parkinson's disease, consistent with shared mitochondrial and protein aggregation mechanisms across these conditions.

The microarray GO and KEGG enrichment results are complementary, with vesicle-mediated transport in synapse and synaptic vesicle cycle emerging as dominant themes. These pathways collectively reflect disruption of synaptic vesicle trafficking, neurotransmitter release, and postsynaptic receptor cycling — processes that are mechanistically linked to synaptic loss, the neuropathological correlate most strongly

associated with cognitive decline in AD. GABAergic synapse dysfunction, also enriched in the microarray dataset, aligns with reports of selective GABAergic interneuron vulnerability in AD temporal cortex.

Taken together, the pathway-level findings suggest that RNA-seq and microarray are capturing complementary facets of the same underlying disease process: energy failure and mitochondrial dysfunction are more prominently detected in the smaller LTL RNA-seq cohort, while synaptic vesicle trafficking impairment dominates the larger MTG microarray dataset. This complementarity argues for the value of multi-platform, multi-region transcriptomic approaches in AD research.

4.6 Machine Learning Classification: Platform-Specific Performance and Feature Importance

Both RNA-seq and microarray datasets yielded robust machine learning classifiers capable of distinguishing AD from control samples. The RNA-seq LASSO logistic regression achieved 100% accuracy under leave-one-out cross-validation, a remarkable result given the small sample size (n = 22). While this performance must be interpreted cautiously — LOOCV on small samples can yield optimistic estimates — the consistent high performance of LASSO across sensitivity and specificity metrics, combined with biologically interpretable feature selection (prioritising VGF and RPH3A), supports genuine discriminative signal rather than overfitting.

The LASSO classifier's reliance on *VGF* (the dominant Random Forest importance feature at MeanDecreaseGini = 0.68) is particularly compelling. VGF's status as the most significant DE gene and the most informative ML feature in the same dataset provides convergent evidence for its central role in LTL-specific AD transcriptomics. *RPH3A* (rabphilin-3A), the second most important RF feature (MeanDecreaseGini = 0.52), encodes a synaptic vesicle-associated protein involved in regulated exocytosis, linking the ML findings to the broader theme of synaptic dysfunction.

For the microarray dataset, the LASSO classifier with 22 selected features achieved 86.8% accuracy on a held-out 20% test set. The top RF importance features — *TNIP1* (immune regulation), *SETD2* (histone H3 lysine 36 methyltransferase; epigenetic regulation), *KATNIP* (microtubule dynamics), and *NEXN* (actin cytoskeleton) — reveal an immune-epigenetic-cytoskeletal axis as the primary discriminative axis in the MTG microarray classifier. The prominence of *SETD2* is particularly noteworthy given emerging evidence for epigenetic dysregulation in AD and the potential for histone methylation patterns to serve as disease biomarkers.

The consistency of LASSO as the top-performing classifier across both datasets — despite differences in validation strategy and sample sizes — suggests that sparse linear models with regularisation are robustly suited to transcriptomic AD classification, outperforming more complex non-linear models that may require larger samples to realise their full discriminative capacity.

4.7 Limitations

Several limitations warrant acknowledgement. First, both datasets rely on post-mortem brain tissue, introducing potential confounders including post-mortem interval and RNA integrity variation. Second, bulk transcriptomic measurements represent population-averaged signals across heterogeneous cell mixtures, obscuring cell-type-specific changes that single-nucleus approaches would resolve. Third, the small RNA-seq cohort (n = 22) limits statistical power and may render ML performance estimates less reliable. Fourth, the cross-platform comparison is confounded by regional differences (LTL vs. MTG), making it impossible to fully disentangle platform effects from biological region effects without a matched multi-region dataset. Fifth, pathway enrichment analysis depends on gene set databases that may not fully represent the brain-specific regulatory landscape.

Despite these limitations, convergent findings across two independent datasets, two platforms, two brain regions, and two independent analytical frameworks provide a robust foundation for the conclusions drawn.

Chapter 5: Conclusion

This thesis presented a comprehensive, two-experiment transcriptomic investigation of Alzheimer′s disease, integrating differential gene expression, pathway enrichment, and machine learning classification across two independent datasets, two sequencing platforms, and two anatomically distinct temporal cortical regions.

Experiment 1 (RNA-seq, GSE104704, Lateral Temporal Lobe) identified 461 significantly differentially expressed genes, with the neuropeptide-encoding gene *VGF* as the most prominent downregulated hit, complemented by *ADCYAP1* and *PCSK1*. Pathway analysis revealed enrichment of purine nucleotide metabolism and core neurodegenerative disease pathways. Machine learning classification achieved 100% accuracy under leave-one-out cross-validation using LASSO logistic regression, with *VGF* and *RPH3A* as the top discriminative features.

Experiment 2 (Microarray, GSE132903, Middle Temporal Gyrus) identified 939 significantly differentially expressed genes from a substantially larger cohort (n = 195), with *RGS4*, *CHGB*, and *SCG2* among the top hits. Pathway analysis highlighted synaptic vesicle cycle dysfunction and GABAergic synapse disruption. LASSO classification achieved 86.8% accuracy on a held-out test set, with *TNIP1*, *SETD2*, *KATNIP*, and *NEXN* as the top discriminative features, pointing to an immune-epigenetic-cytoskeletal axis of dysregulation.

The cross-platform comparative study revealed 47 overlapping significant genes between datasets, of which 46 displayed direction reversal — yielding a strongly negative Pearson fold-change correlation ($r = -0.438$). This finding is interpreted as genuine regional transcriptomic heterogeneity between the LTL and MTG, rather than analytical artefact, and underscores the importance of brain-region specificity in transcriptomic AD research. The single concordantly dysregulated gene across both platforms, *ANGPTL4*, was consistently downregulated and represents a high-priority cross-regional AD biomarker candidate.

Key Contributions

This study makes several important contributions to the AD transcriptomics literature. First, it demonstrates that LASSO logistic regression is a robust classifier for AD transcriptomic data across both small-sample RNA-seq (LOOCV) and large-sample microarray (80/20 split) settings. Second, it identifies *ANGPTL4* as a cross-platform, cross-region concordant AD candidate warranting independent validation. Third, it characterises the direction reversal phenomenon as biologically meaningful, arguing for region-matched experimental designs in future multi-platform AD transcriptomic studies. Fourth, it provides fully reproducible analytical pipelines (DESeq2/ashr for RNA-seq, limma/eBayes for microarray, clusterProfiler for enrichment, and R-based ML classifiers) that can be directly applied to future AD datasets.

Future Directions

Several avenues of future work flow naturally from these findings. Validation of *ANGPTL4* downregulation in independent post-mortem cohorts spanning multiple brain regions would establish its candidacy as a region-agnostic AD biomarker. Single-nucleus RNA sequencing of LTL and MTG tissue from the same individuals would resolve cell-type-specific expression patterns underlying the bulk transcriptomic direction reversal. Protein-level validation of key ML features (VGF, TNIP1, SETD2) by immunohistochemistry or mass spectrometry-based proteomics would bridge the transcriptomic findings to

disease-relevant protein changes. Finally, integration of the transcriptomic findings with genetic data (eQTL analysis, colocalization with GWAS loci) would position identified genes within the broader AD genetic risk architecture.

In conclusion, this thesis demonstrates the power and complementarity of multi-platform transcriptomic analysis for characterising the molecular heterogeneity of Alzheimer's disease, while highlighting the critical role of brain-region specificity in interpreting cross-dataset comparisons. The findings advance our understanding of the transcriptomic basis of AD and provide a foundation for biomarker discovery and therapeutic target identification.

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